

Project Report: Enterprise Document Intelligence Platform - Generative AI Engineer

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Family:	Generative AI Engineer
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Multi-tenant RAG with hybrid search, reranking, guardrails, and admin analytics dashboard.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview** *(Level: Advanced)

Enterprise Document Intelligence Platform** is a advanced-level end-to-end project for **Generative AI Engineer**. **Duration:** 5-6 weeks. **Multi-tenant RAG with hybrid search, reranking, guardrails, and admin analytics dashboard.** **Tools:** LangChain, Elasticsearch, OpenAI, React.

Code Reference:

Objective: Hybrid search (BM25 + vector)
Objective: Cross-encoder reranking
Objective: PII redaction guardrails
Objective: Admin dashboard with usage analytics

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Generative AI Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Generative AI Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Hybrid [Advanced]

Chapter focus: Phase: Hybrid** *(Level: Advanced)

BM25 + vector fusion. Complete these tasks: Reciprocal rank fusion; Tuning. Deliverable for this phase: Search quality benchmark.

Code Reference:

- [] Reciprocal rank fusion
- [] Tuning

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Hybrid is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Search quality benchmark to the teacher.

Chapter 3: Phase: Rerank [Advanced]

Chapter focus: Phase: Rerank** *(Level: Advanced)

Cross-encoder reranker. Complete these tasks: Top-20 ? Top-5; Latency test. Deliverable for this phase: Reranker eval.

Code Reference:

- [] Top-20 ? Top-5
- [] Latency test

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Rerank is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Reranker eval to the teacher.

Chapter 4: Phase: Guardrails [Advanced]**Chapter focus: Phase: Guardrails** *(Level: Advanced)**

PII filter + topic block. Complete these tasks: Regex + LLM check; Audit log. Deliverable for this phase: Safety test cases.

Code Reference:

- ☐ Regex + LLM check
- ☐ Audit log

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Guardrails is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Safety test cases to the teacher.

Chapter 5: Phase: Admin [Advanced]**Chapter focus: Phase: Admin** *(Level: Advanced)**

Usage dashboard. Complete these tasks: Token costs; Query logs. Deliverable for this phase: Admin UI screenshot.

Code Reference:

- ☐ Token costs
- ☐ Query logs

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Admin is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Admin UI screenshot to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review *(Level: Advanced)**

Submit all final deliverables: Platform demo; Search benchmark; Safety test report; Admin dashboard. Skills demonstrated: Hybrid search, Safety, Enterprise RAG. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Platform demo
Deliverable: Search benchmark
Deliverable: Safety test report
Deliverable: Admin dashboard
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide**What it actually is**

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Platform demo and passes the rubric.